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FINAL PROJECT PRESENTATION

MACHINE LEARNING PROBLEM

Goal: The task is to predict child stunting prevalence using socio-economic and demographic indicators such as age, sex, household wealth, education level, and place of residence.

Stunting reflects chronic undernutrition and long-term health outcomes

Driven by socio-economic and demographic factors
Supports evidence-based public health interventions

Type: Regression because the target variable, stunting prevalence, is continuous rather than categorical.

DATA SET DESCRIPTION

- Dataset: Child Malnutrition (UNICEF/WHO/World Bank)
- Source: WHO Health Inequality Data Repository (https://datasafe-h5afbhf4gwctabaa.z01.azurefd.net/api/Download/TOP/rep_nut/data)
- Rows: 91,571
- Columns: 24
- Global coverage with disaggregated indicators

KEY VARIABLES

Target: Stunting prevalence

Predictors:

- ❑ Age group, sex
- ❑ Wealth status, maternal education
- ❑ Place of residence, region
- ❑ Country and year

Approach

- Preprocessing: Handle missing data, encode categories
- Models: Linear Regression, Neural Network
- Evaluation: Root Mean Squared Error (RMSE), R^2(coefficient of determination)
- Goal: Compare model performance and identify key predictors

Modelling and Evaluation

- Two models were applied: Linear Regression and a neural network (Multilayer Perceptron). Model performance was evaluated using RMSE and R^2.

Results

- Neural Network RMSE (12.17): On average, the model's predictions deviate from the actual values by approximately 12.17 units.
- Linear Regression (14.94): The predictions are less accurate, deviating by 14.94 units on average.

Overall Performance Comparison

- Neural Network is superior: The Neural Network outperforms the Linear Regression model on both metrics. It has a lower error (RMSE) and explains more variance in the data coefficient of determination). The neural network outperformed the linear regression model, achieving a lower RMSE (12.17 vs 14.94) and higher R^2 (0.39 vs 0.06). This indicates that the neural network was better able to capture the underlying patterns in the data.
- Both models are relatively weak: While the Neural Network is better, an of 0.39 indicates it only explains 39% of the variance.

Results Interpretation:

- The results suggest that complex, non-linear relationships between socio-economic and demographic factors influence child stunting prevalence. Income group and population characteristics appear to contribute to variation in malnutrition outcomes. However, model performance is moderate, indicating that additional predictors could improve accuracy.
- Income group and region appear to be strong determinants of malnutrition.
- The findings emphasize that more advanced models, such as neural networks, are better suited for capturing complex relationships in real-world health data.

Conclusions

- This project demonstrates the application of machine learning in public health analytics. By combining data exploration with predictive modelling, it highlights the importance of socio-economic factors in shaping child malnutrition outcomes.
- The findings emphasize that more advanced models, such as neural networks, are better suited for capturing complex relationships in real-world health data.